

CLUJ NAPOCA, Romania

WMO: 151200

Lat: 46.7778N

Lon: 23.5714E

Elev: 1349

StdP: 13.99

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.7	12.0	2.8	6.7	9.5	7.3	8.4	13.8	19.8	31.0	16.3	31.9	2.6	50	0.482

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.6	88.5	68.8	85.7	68.1	82.5	66.5	71.4	83.6	69.7	81.7	68.1	79.2	5.7	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.6	106.8	76.3	66.0	100.7	74.2	64.4	95.2	72.2	36.0	83.8	34.5	81.5	33.2	79.0	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.2	13.1	10.8	DB	-0.1	93.3	6.3	3.8	-4.6	96.1	-8.2	98.3	-11.8	100.4	-16.3	103.1	(4)
(5)			WB	-0.3	74.7	6.2	1.9	-4.8	76.1	-8.4	77.2	-11.9	78.3	-16.4	79.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.6	27.9	32.2	40.6	51.0	59.4	65.7	68.7	68.5	59.8	49.8	39.9	30.1		
	DBStd	16.14	8.29	8.88	8.15	6.91	6.15	5.79	5.13	5.30	6.39	7.27	8.01	8.33		
	HDD50	2591	684	499	303	69	6	0	0	0	4	95	314	617		
	HDD65	6030	1149	919	758	420	189	60	24	26	178	472	753	1082		
	CDD50	2429	0	0	10	100	298	472	579	572	297	89	12	0		
	CDD65	392	0	0	0	0	16	82	138	133	22	1	0	0		
	CDH74	4123	0	0	0	31	259	834	1357	1348	282	12	0	0		
	CDH80	1243	0	0	0	3	35	224	440	477	63	1	0	0		
(14) Wind	WSAvg	4.3	3.6	4.1	5.0	5.2	4.8	4.9	4.5	4.3	4.2	3.9	3.8	3.5		
(15) Precipitation	PrecAvg	24.7	1.2	1.0	1.3	2.0	2.8	3.6	3.4	2.5	2.2	1.8	1.5	1.5		
	PrecMax	32.1	2.1	2.3	3.8	4.8	5.4	7.3	6.8	6.8	5.3	4.0	3.5	2.8		
	PrecMin	15.8	0.4	0.2	0.2	0.3	1.1	0.6	0.7	0.1	0.2	0.1	0.0	0.2		
	PrecStd	4.8	0.5	0.5	0.9	1.0	1.2	1.5	1.5	1.6	1.2	1.1	0.7	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.9	55.8	67.6	77.9	84.0	89.3	92.2	93.5	86.5	75.2	64.6	51.5		
		MCWB	44.5	47.0	52.5	59.5	63.2	70.3	69.5	68.9	65.1	60.3	54.8	46.0		
	2%	DB	44.6	51.4	62.4	72.7	79.9	85.7	88.1	89.2	80.8	70.9	60.2	46.7		
		MCWB	41.4	44.9	49.3	57.0	62.8	68.4	69.2	68.7	63.2	58.5	52.5	43.7		
	5%	DB	41.2	47.4	57.7	68.3	76.9	82.4	84.8	85.6	76.9	66.5	56.3	43.2		
		MCWB	38.7	42.7	47.0	54.1	61.7	66.7	68.2	68.0	62.2	56.8	50.4	41.1		
	10%	DB	38.1	43.9	53.8	64.6	73.3	79.0	81.7	81.7	73.0	62.2	52.0	40.6		
		MCWB	36.3	40.5	45.3	52.4	60.4	65.6	67.1	66.4	60.5	54.6	47.9	39.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.9	48.5	53.6	61.2	67.3	73.4	73.4	73.3	68.8	62.8	55.9	47.0		
		MCDB	48.2	53.5	64.0	74.4	78.5	85.1	84.7	86.1	80.3	71.2	62.2	49.8		
	2%	WB	41.8	45.5	50.6	58.3	64.6	70.4	71.2	71.0	65.5	60.0	53.3	43.7		
		MCDB	44.1	50.2	59.7	70.7	75.8	81.5	83.0	83.8	76.0	67.8	58.7	46.3		
	5%	WB	39.0	43.2	48.3	55.7	62.8	68.3	69.6	69.2	63.6	57.7	50.8	41.4		
		MCDB	40.9	46.9	56.2	65.4	73.7	78.6	81.8	81.4	73.1	64.9	55.5	43.1		
	10%	WB	36.5	40.6	46.1	53.6	61.2	66.6	68.1	67.5	61.8	55.5	48.1	39.1		
		MCDB	38.0	43.6	53.0	61.8	71.3	76.8	79.2	78.6	70.1	60.9	52.0	40.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.0	13.8	17.9	20.7	21.3	21.0	21.6	22.8	20.8	19.6	14.4	10.1		
		MCDBR	13.4	18.2	25.4	27.2	27.8	26.6	27.2	28.8	27.7	26.2	21.0	13.4		
		MCWBR	10.1	12.4	14.4	14.1	12.9	11.9	11.5	11.1	12.1	14.8	13.6	10.3		
	5% WB	MCDBR	12.7	17.2	23.1	24.3	24.9	24.0	24.4	25.3	23.8	23.6	18.9	12.9		
		MCWBR	9.9	12.3	13.5	13.4	12.5	11.8	11.4	11.1	11.7	13.9	12.8	10.3		
(40) Clear-Sky Solar Irradiance	taub		0.312	0.339	0.381	0.426	0.412	0.408	0.427	0.431	0.393	0.365	0.339	0.313		
	taud		2.336	2.290	2.229	2.144	2.245	2.285	2.239	2.226	2.303	2.359	2.380	2.365		
	Ebn at Noon		244	259	264	262	271	272	265	258	258	247	231	229		
	Edh at Noon		24	30	38	45	42	41	42	41	35	28	23	21		
(44) All-Sky Solar Radiation	RadAvg		371	626	998	1386	1681	1848	1812	1670	1181	797	452	287		
	RadStd		52	83	144	142	163	147	160	156	136	97	62	39		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.44	N/A	N/A	+2.11	N/A	N/A	-320	-450	+241	+99
(47) Regional (64 neighbors)		+1.28	N/A	N/A	+1.69	N/A	N/A	-256	-376	+189	+67

Nomenclature: See separate page